AI-Powered Full Stack Developer Program

ANSWER KEYS · INSTRUCTOR ONLY

Week 2 · Day 2

Semantic HTML, Accessibility, Media → CSS Basics

Every bug-hunt's BUGS.md, the other answer files, and the HW3 answer key

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Stage 1, AI as Tutor: AI may explain; you type every line.

Tech Skills Institute (TSI) Nepal · Kathmandu

D2-00-outfit-swap · README-answer.md

D2-00 · Outfit swap — answer

File: classwork/D2-00-outfit-swap/solution/README-answer.md

1. **The only change** is the `href` on line 7:
   - `css/outfit-traditional.css` → cream paper, cherry headings, Georgia (serif)
   - `css/outfit-modern.css` → navy band, white cards, round red nav buttons, Segoe UI / Arial (sans-serif)
   - `css/outfit-night.css` → dark background, gold headings, Courier New (monospace)
2. **Did the `<body>` change?** No. Not one character of the HTML changed. This is the CSS Zen Garden idea: HTML is the skeleton (content and structure), CSS is the outfit (colours, fonts, spacing).
3. **A selector in all three files:** `body`, `h1`, `h2`, `nav a`, `main` and `footer` are in all three. The selector is the same ("who"), the declarations are different ("what").
4. **Why this matters:** one stylesheet can dress 100 pages (the school uniform, B4 and D2-15). Change one file and the whole site changes.

D2-02-semantic-misuse-bughunt · BUGS.md

D2-02 Semantic misuse — answer key (BUGS.md)

File: classwork/D2-02-semantic-misuse-bughunt/solution/BUGS.md

Page: "TSI Blog". No JavaScript in this exercise: the button only has to be a real, focusable button.

Line = the line in the **starter** file. Tags: V = see it · D(HH) = HTMLHint · D(F12) = Elements panel · D(LH) = Lighthouse · D(W3C) = W3C validator · R = rule check.

#	File	Line	Bug	Fix	Tag
1	`index.html`	31	Two `<main>` elements. A page has only **one** main content area.	Delete the second `<main>` and its `</main>`. Move its `<section>` into the first `<main>`.	R / D(W3C)
2	`index.html`	11	`<nav>` wraps a paragraph of text. `<nav>` is only for a group of navigation **links**.	Remove `<nav>` and `</nav>`; keep the `<p>`.	R
3	`index.html`	32	`<section>` with no heading. Every section needs a heading that says what it is.	Add `<h2>Get new posts</h2>` as the first line inside the section.	R
4	`index.html`	23, 16	Swapped: the main blog post sits in `<aside>`, and the "Related posts" box sits in `<main>`.	Put the `<article>` inside `<main>`. Put "Related posts" (h2 + ul) inside `<aside>`.	R
5	`index.html`	34	`<div class="btn">Subscribe</div>` is not a button: it looks like plain text and the **Tab** key skips it.	`<button type="button">Subscribe</button>`	V

Reveal tip: press Tab through the starter page on the projector. Focus jumps over "Subscribe". Then press Tab on the solution: the button gets a focus ring.

D2-03-a11y-audit · BUGS.md

D2-03 A11y audit: TSI Admissions — answer key (BUGS.md)

File: classwork/D2-03-a11y-audit/solution/BUGS.md

Page: "TSI Admissions". Each bug counts once (bug 5 is one bug in three places).

Line = the line in the **starter** file. Tags: V = see it · D(HH) = HTMLHint · D(F12) = Elements panel · D(LH) = Lighthouse · D(W3C) = W3C validator · R = rule check.

#	File	Line	Bug	Fix	Tag
1	`index.html`	2	`<html>` has no `lang`. A screen reader does not know which language to speak.	`<html lang="en">`	D(LH) / D(HH)
2	`index.html`	6	`<title>Document</title>` is the Emmet default. It tells nobody what the page is (tab, bookmark, Google result).	`<title>TSI Admissions · Tech Skills Institute, Kathmandu</title>`	R
3	`index.html`	22	The TSI building photo has no `alt`. A blind visitor hears only "image" (or the file name).	Add a meaningful alt: `alt="The TSI building in Dillibazar: …"`	D(LH) / D(HH)
4	`index.html`	15	The decorative dhaka strip has a long alt. A screen reader reads all of it, and it means nothing.	Decoration gets an **empty** alt: `alt=""`	R
5	`index.html`	24, 43, 55	Three links say "click here" (one bug, three places). Link text must say where it goes.	"See the course fees" · "Go to the application form" · "Read about our campus"	R
6	`index.html`	52	The phone input has only a placeholder and no `<label>`. The hint disappears as soon as you type, and clicking the text does not focus the box.	Add `<label for="phone">Phone number</label>` before the input.	V

Where Lighthouse shows them (F12 → Lighthouse → tick Accessibility only → Analyze page load):

- Bug 1 → Accessibility: "`<html>` element does not have a `[lang]` attribute".
- Bug 3 → Accessibility: "Image elements do not have `[alt]` attributes".
- Bugs 2, 4 and 5 are rule checks (R): read the code. (Lighthouse's "link text" check skips links to `#` on the same page, so it does not catch these "click here" links. On a link to another page it would.)
- Bug 6 is V: click the words "Phone number" (nothing happens) and type in the box (the hint disappears). Lighthouse accepts a placeholder as a label, so it stays quiet: a good reason not to trust one tool only.

Reveal tip: turn on Narrator (Ctrl+Win+Enter) and Tab through the starter, then through the solution.

D2-04-broken-media · BUGS.md

D2-04 Broken media — answer key (BUGS.md)

File: classwork/D2-04-broken-media/solution/BUGS.md

Page: "TSI Canteen Media Corner" (`index.html`) + the map page shown in the iframe (`kathmandu-map.html`, no bugs).

Line = the line in the **starter** file. Tags: V = see it · D(HH) = HTMLHint · D(F12) = Elements panel · D(LH) = Lighthouse · D(W3C) = W3C validator · R = rule check.

#	File	Line	Bug	Fix	Tag
1	`index.html`	17	The first `<audio>` has no `controls`, so nothing shows on the page.	`<audio src="media/canteen-bell.wav" controls></audio>`	V
2	`index.html`	19	The second audio points to `media/canteen_bell.wav` (underscore). The file is `canteen-bell.wav` (dash). The player shows 0:00 and will not play.	`src="media/canteen-bell.wav"`	V
3	`index.html`	23	The `<video>` has no `controls`: no play button, so nobody can start it.	Add `controls` to `<video>`.	V
4	`index.html`	23	`poster="image/poster-momo-steam.svg"`: the folder is `images`. Before play you see an empty box, not the poster.	`poster="images/poster-momo-steam.svg"`	V
5	`index.html`	30	`<iframe src="kathmandu-map.htm">`: the file is `.html`. The window shows a 404 page.	`src="kathmandu-map.html"`	V

Reveal tip: fix bug 3 before bug 4 on the projector. With controls and no poster, the room sees an empty box with a play button. Then fix the poster path and the momo picture appears.

D2-06-stylesheet-not-loading · BUGS.md

D2-06 Stylesheet not loading — answer key (BUGS.md)

File: classwork/D2-06-stylesheet-not-loading/solution/BUGS.md

Page: "Meet Didi" (`index.html` + `css/style.css`). Bugs 1 and 2 are on the same line (the `<link>`).

Line = the line in the **starter** file. Tags: V = see it · D(HH) = HTMLHint · D(F12) = Elements panel · D(LH) = Lighthouse · D(W3C) = W3C validator · R = rule check.

#	File	Line	Bug	Fix	Tag
1	`index.html`	7	`rel="stylsheet"` (typo). The browser does not know this is a stylesheet, so **no CSS at all** loads.	`rel="stylesheet"`	V
2	`index.html`	7	`href="css/styles.css"`, but the file is `css/style.css` (no **s**). 404: no CSS at all.	`href="css/style.css"`	V
3	`css/style.css`	32	The `h1` rule has no closing `}`. The browser keeps reading and swallows the next rule: `.highlight` never works (the yellow marker on "She still folds every momo by hand" is missing).	Add `}` after `font-size: 3rem;`.	V
4	`css/style.css`	15	Missing `;` after `color: #2a3f63`. The browser reads `color: #2a3f63 line-height: 1.8` as one broken value and drops it: the text is not navy and the lines are tight. (Both declarations are lost, not only the second.)	`color: #2a3f63;`	V
5	`css/style.css`	10	`colour: maroon;`: British spelling. CSS only knows `color`, so the h2 text stays black (the line under it is maroon).	`color: maroon;`	V
6	`css/style.css`	20	`// footer styles` is a JavaScript comment. In CSS it becomes part of the next selector, so the whole `footer` rule is dropped: no navy band.	`/* footer styles */`	V

Reveal tip: after bugs 1 + 2, open F12 → Sources (or Network) to show `style.css` now loads. Then F12 → Elements → click `<h2>`: the Styles panel shows `colour` crossed out with a ⚠ "Invalid property name". That is the CSS spell-checker.

About bug 3: the `h1` rule itself still works. Modern Chrome reads the swallowed `.highlight { … }` as a rule *nested inside* `h1`, so it would only style a `.highlight` inside the h1. There is none, so the marker disappears.

D2-08-selector-mixups · BUGS.md

D2-08 Selector mix-ups — answer key (BUGS.md)

File: classwork/D2-08-selector-mixups/solution/BUGS.md

Page: "Momo House Menu" (`index.html` + `css/style.css`).

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#	File	Line	Bug	Fix	Tag
1	`index.html`	25	HTML `class="Card"` but the CSS says `.card`. Class names are case-sensitive, so the Fried momo card has no border or white box.	`class="card"`	V
2	`css/style.css`	15	CSS `#menu`, but the HTML is `id="main-menu"`. Nothing matches, so the nav has no navy band (the white links become invisible on cream).	`#main-menu {`	V
3	`css/style.css`	33	`price { … }` has no dot, so it looks for a `<price>` tag. There is none: prices are not big, bold and red.	`.price {`	V
4	`css/style.css`	10	`h1, h2, {` has a trailing comma. One bad part of a selector list drops the **whole** rule: no cherry Georgia headings.	`h1, h2 {`	V
5	`css/style.css`	39	`:hoverr` (typo). An unknown pseudo-class makes the selector invalid, so the "Order" links never change on hover.	`.card a:hover {`	V (on hover)

Reveal tip: in F12 → Elements, click the Fried card. The Styles panel shows no `.card` rule at all: the selector did not match. That is how you spot a selector mistake (versus a crossed-out declaration).

D2-10-units-and-colours · BUGS.md

D2-10 Units & colours — answer key (BUGS.md)

File: classwork/D2-10-units-and-colours/solution/BUGS.md

Page: "TSI Canteen Prices" (`index.html` + `css/style.css`).

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#	File	Line	Bug	Fix	Tag
1	`css/style.css`	16	`width: 300;` has no unit. The browser drops it, so the price board stretches across the whole page.	`width: 300px;`	V
2	`css/style.css`	23	`#gg0000` is not a hex colour (hex uses only 0–9 and a–f). Dropped: prices are navy, not red.	`color: #cc0000;`	V
3	`css/style.css`	28	`font-size: 1.5 rem;` has a space between the number and the unit. Dropped: the special is not bigger.	`font-size: 1.5rem;`	V
4	`css/style.css`	3	`font-family: "Momo Sans";`: that font does not exist on any computer, and there is no fallback, so the browser uses Times New Roman.	`font-family: "Momo Sans", Arial, sans-serif;`	V
5	`css/style.css`	34	`line-height: 1.5px;` makes each line only 1.5 pixels tall, so the lines of the note sit on top of each other.	`line-height: 1.5;` (no unit = 1.5 × the font size)	V

Reveal tip: F12 → Elements → click `.price-board`. In Styles, `width: 300;` is crossed out with ⚠ "Invalid property value". Bugs 1, 2 and 3 all show this way. Bugs 4 and 5 are valid CSS: no warning, only a bad result.

D2-12-broken-cards · BUGS.md

D2-12 Broken cards — answer key (BUGS.md)

File: classwork/D2-12-broken-cards/solution/BUGS.md

Page: "TSI Morning Batch" (`index.html` + `css/style.css`). Open it in a wide window (1280 px or more).

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#	File	Line	Bug	Fix	Tag
1	`css/style.css`	27	Each card is 300 + 2×20 padding + 2×5 border = **350 px** wide (content-box). Three cards = 1050 px, but `.row` is 960 px, so the third card wraps to a new line.	Add `box-sizing: border-box;` to `.card`: now 300 px *includes* padding and border.	V
2	`css/style.css`	10	`.wrapper { margin: 0 auto; }` has no width, so it is already full width and cannot centre: the cream box fills the whole window.	Add `max-width: 1000px;` (or `width: 1000px;`).	V
3	`css/style.css`	43	`padding: -10px;`: padding can never be negative. The browser drops it, so the red "Morning batch" tag is squashed against its text.	`padding: 4px 10px;`	V
4	`css/style.css`	16	`border: 2px orange;` has no style. The default style is `none`, so there is no border around the note.	`border: 2px solid orange;`	V
5	`css/style.css`	36	The avatar is 120 × 80. `border-radius: 50%` on a rectangle makes an **oval**, not a circle.	Equal width and height: `height: 120px;`	V

Reveal tip: F12 → Elements → hover a card. The box diagram in **Computed** shows 300 + 40 + 10 = 350. Add `box-sizing: border-box` live in the Styles panel and watch the third card jump up.

D2-14-nav-wont-behave · BUGS.md

D2-14 Nav won't behave — answer key (BUGS.md)

File: classwork/D2-14-nav-wont-behave/solution/BUGS.md

Page: Momo House header and hero (`index.html` + `css/style.css`).

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#	File	Line	Bug	Fix	Tag
1	`css/style.css`	43	`display: blok;` (typo). Dropped, so "Order now" stays an inline link: `width: 200px` and the top margin are ignored, so it is a small button squeezed against the text above.	`display: block;`	V
2	`css/style.css`	15	`.logo { display: none; }` was left in by mistake. The logo has left the room: it is not on the page at all.	Delete `display: none;` from `.logo`.	V
3	`css/style.css`	18	`.menu` has no `list-style: none`, so the menu shows bullets. (You see them while the `li`s are still blocks, bug 5. Mend 3 and 5 together.)	Add `list-style: none;` to `.menu`.	V
4	`css/style.css`	28	Nav links are `display: inline` with `padding: 12px 16px; margin-top: 20px;`. Inline boxes ignore vertical margin, and their vertical padding overlaps the boxes above and below (here: into the cream hero).	`display: inline-block;` (side by side, but keeps its own size and margin)	V
5	`css/style.css`	23	CSS `.menu-li`, but the HTML is `<ul class="menu"><li>`. No element has the class `menu-li`, so the `li`s stay blocks: the menu is a vertical list, not a bar.	`.menu li {` (a space: the `li`s inside `.menu`)	V

Reveal tip: use the queue and the bench. Block = the queue (each on its own line). Inline = the bench (side by side, you can't push them up or down). Inline-block = side by side, but each keeps its own size. None = left the room.

D2-FF3-specificity-duel · ANSWERS.md

D2-FF3 · Specificity duel — answers

File: classwork/fast-finishers/D2-FF3-specificity-duel/solution/ANSWERS.md

Score: id = (1,0,0) · class / pseudo-class = (0,1,0) · element = (0,0,1). Compare from the left. A tie → the rule written **later** wins. An inline `style=""` beats every selector. A rule that targets the element beats any colour it would inherit from its parent.

#	Element	Rules in the fight	Winner	Colour	Why
1	`<p class="note">`	`p` (0,0,1) vs `.note` (0,1,0)	`.note`	**green**	class beats element
2	`<p class="note" id="p2">`	`p`, `.note` vs `#p2` (1,0,0)	`#p2`	**red**	id beats class
3	`<p class="twice">`	`.twice` vs `.twice` (both 0,1,0)	the second `.twice`	**purple**	a tie: the later rule wins
4	`<p class="p4">` inside a div	`.p4` (0,1,0) vs `div p` (0,0,2)	`.p4`	**teal**	one class beats two elements, even though `div p` is written later
5	`<p class="a b">`	`.a.b` (0,2,0) vs `.b` (0,1,0)	`.a.b`	**crimson**	two classes beat one, even though `.b` is later
6	`<p id="p6" style="color: navy;">`	`#p6` vs inline style	inline	**navy**	an inline style beats every selector (that is why we avoid inline styles)
7	`<span>` inside `<p class="parent">`	inherited red vs `span` (0,0,1)	`span`	**green**	a rule that targets the span beats a colour inherited from the parent
8	`<h2 class="title big">` inside `#duel`	`#duel h2` (1,0,1) vs `.title.big` (0,2,0)	`#duel h2`	**maroon**	one id beats any number of classes

Check in DevTools: F12 → Elements → click the element. In Styles, the winning declaration is at the top and the losers are ~~crossed out~~ below it.

D2-FF4-haunted-stylesheet · BUGS.md

D2-FF4 Haunted stylesheet — answer key (BUGS.md)

File: classwork/fast-finishers/D2-FF4-haunted-stylesheet/solution/BUGS.md

Page: "Kathmandu Ghost Walk". All 10 mistakes are in `css/style.css`.

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#	File	Line	Bug	Fix	Tag
1	`css/style.css`	77	Missing `}` after the `.warning` rule. The next rule, `.highlight`, is swallowed: "no bell in the courtyard" has no green marker.	Add `}` after `font-weight: bold;` in `.warning`.	V
2	`css/style.css`	19	`padding: 20;` has no unit. Dropped: the header has no space around it.	`padding: 40px;`	V
3	`css/style.css`	23	CSS `.Title`, but the HTML is `class="title"`. Case matters: the h1 is not ghost-green or big.	`.title {`	V
4	`css/style.css`	13	CSS `#nav`, but the HTML is `id="navbar"`. The nav bar has no purple background.	`#navbar {`	V
5	`css/style.css`	42	`colour` is not a CSS property. The h2 headings stay grey instead of gold.	`color: #f5b700;`	V
6	`css/style.css`	64	Missing `;` after `color: #8a8a8a`. Both declarations are lost: the footer is not grey and not centred.	`color: #8a8a8a;`	V
7	`css/style.css`	36	`border: 1px red;` has no style, so the story boxes have no border.	`border: 1px solid red;`	V
8	`css/style.css`	52	`display: blok;` (typo). The badge stays inline, so `width: 140px` is ignored.	`display: block;`	V
9	`css/style.css`	31	`font-size: 24 px;` has a space between number and unit. Dropped: the subtitle is normal size.	`font-size: 24px;`	V
10	`css/style.css`	69	`// rules list` is a JavaScript comment. It breaks the next selector, so the rules list is not green.	`/* rules list */`	V

D2-FF5-lost-and-found-bughunt · BUGS.md

D2-FF5 Lost & Found — answer key (BUGS.md)

File: classwork/fast-finishers/D2-FF5-lost-and-found-bughunt/solution/BUGS.md

Page: "TSI Open Day" (`index.html` + `css/style.css`; `kathmandu-map.html` has no bugs). These are the Day 2 mistakes that did not fit in the class hunts: semantic (1–3), a11y (4–7), stylesheet (8–9), selectors (10–11).

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#	File	Line	Bug	Fix	Tag
1	`index.html`	56	A `<header>` inside the `<footer>`. A footer cannot contain a header.	Remove the `<header>` and `</header>` lines; keep the h2.	D(W3C)
2	`index.html`	12	`<article>` used for the site navigation bar. An article is a stand-alone story; links to other parts of the site go in `<nav>`.	`<nav> … </nav>`	R
3	`index.html`	27	`<b>` for "Warning:". `<b>` only looks bold; `<strong>` says "this is important".	`<strong>Warning:</strong>`	R
4	`index.html`	21	Heading order h1 → h3 → h2. Headings must not skip a level.	`<h2>Programme</h2>`	D(LH)
5	`index.html`	45	`<div class="btn primary">Apply</div>` is not a button: Tab skips it.	`<button type="button" class="btn primary">Apply</button>`	V
6	`index.html`	33	The fee table has no `<th>`: the first row is `<td>` with `<b>`. A screen reader does not know these are headers.	`<th scope="col">Course</th>` and `<th scope="col">Fee</th>`	R
7	`index.html`	52	The `<iframe>` has no `title`. A screen reader says only "frame".	Add `title="Map of Kathmandu with the TSI campus"`.	D(LH)
8	`css/style.css`	16	`font-size: 20 px;` (space). Dropped: the intro is normal size.	`font-size: 20px;`	V
9	`css/style.css`	5	`background-color = #fdf6ec;` uses `=` instead of `:`. Dropped: the page is white, not cream.	`background-color: #fdf6ec;`	V
10	`css/style.css`	30	`.btn .primary` (with a space) means "a .primary **inside** a .btn". The HTML has both classes on **one** element: `class="btn primary"`.	`.btn.primary {` (no space)	V
11	`index.html`	49	`class="card, special"`: no commas in a class list. The browser sees the classes `card,` and `special`, so `.card` never matches (no white box).	`class="card special"`	V

D2-HW1-haunted-portfolio-repair · BUGS.md

D2-HW1 Haunted portfolio repair — answer key (BUGS.md)

File: homework/_solutions/D2-HW1-haunted-portfolio-repair/BUGS.md

Rajan's 2-page portfolio (`index.html`, `projects.html`, `css/style.css`). Bug 1 is only on the home page; the projects page links the CSS correctly.

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#	File	Line	Bug	Fix	Tag
1	`index.html`	7	`<link href="style.css">`, but the file is `css/style.css`. The home page has no CSS at all.	`href="css/style.css"`	V
2	`projects.html`	31	Two `<main>` elements on the projects page.	Delete the extra `</main>` + `<main>` pair, so the section is inside the one main.	R / D(W3C)
3	`index.html`	24	The second `<section>` has no heading.	Add `<h2>About me</h2>`.	R
4	`index.html`	20	The avatar `<img>` has no `alt`.	`alt="Photo of Rajan, smiling, with round glasses"`	D(HH) / D(LH)
5	`css/style.css`	21	`.nav li { display: inline-blok; }` (typo). The menu stays vertical.	`display: inline-block;`	V
6	`css/style.css`	30	CSS `.Card`, HTML `class="card"`. Case matters: the project cards have no box at all.	`.card {`	V
7	`css/style.css`	11	`.site-header { padding: 20; }` has no unit. Dropped: the menu is squashed against the edge of the navy bar.	`padding: 20px;`	V
8	`css/style.css`	35	`border: 1px #999;` has no style, so the cards have no border.	`border: 1px solid #999;`	V
9	`css/style.css`	33	Each card is 420 + 2×20 + 2×1 = 462 px wide. Two cards do not fit in the 900 px main, so the second card drops below the first.	Add `box-sizing: border-box;` to `.card`.	V
10	`css/style.css`	63	Missing `}` after the `.hero` rule. The next rule, `main`, is lost: the content is not in a centred 900 px column any more (it runs edge to edge).	Add `}` after `border-radius: 12px;`.	V
11	`css/style.css`	25	`color: #fffff;` has 5 hex digits. Dropped: the nav links are blue on navy (hard to read).	`color: #ffffff;` (6 digits) or `white`	V
12	`css/style.css`	52	CSS `#footer`, but the HTML is `<footer class="footer">` (a class, not an id). The footer is plain.	`.footer {`	V

D2-HW3 · Answer Key: Semantic HTML, Accessibility, Media → CSS Basics

File: homework/_solutions/D2-HW3-Answer-Key.md

TSI Nepal · AI-Powered Full Stack Developer Program · Week 2, Day 2 · Instructor copy (share after students try)

Try every question before you read this. Say your answer out loud first; interviews are spoken.

Every CSS answer below was checked in Chrome with `getComputedStyle` and `offsetWidth` (headless, October 2026).

Part 1: Self-check answers

S1. Top of the page → `<header>`. Menu of links → `<nav>`. Main content → `<main>` (only one). A post that makes sense on its own → `<article>`. Side box → `<aside>`. Bottom → `<footer>`. These are **boxes with names**, like rooms in a house: kitchen and bedroom, not "room 1, room 2".

S2. A `<div>` is not a button to the browser. The **Tab key skips it**, Enter does nothing, and a screen reader doesn't say "button". Test: press Tab through the page; the div never gets the focus ring. Use `<button>`. See D2-02 bug 5.

S3. TSI building: `alt="The Tech Skills Institute building in Dillibazar, Kathmandu"`. Decorative strip: `alt=""` (empty), so screen readers skip it. See D2-03 bugs 3 and 4.

S4. Any three: `<html lang="en">` (the reader picks the right voice), `<meta charset="UTF-8">`, `<meta name="viewport" …>` (phones), a descriptive `<title>` (tab, search result), `<meta name="description" …>` (the search result snippet), a favicon `<link rel="icon">`.

S5. `controls` is missing; without it the audio player is invisible. `poster` is the picture shown in the video box **before** the video plays. See D2-04 bugs 1 and 4.

S6. `<link rel="stylesheet" href="css/style.css">` is the **school uniform**: one file dresses every page. Change one line, and all 4 pages change. Inline style = a sticky note on one shirt; `<style>` = an outfit list for one person.

S7. `p` · `.veg` · `#todays-special` · `nav a` · `.dish.spicy` (no space). These are the D2-09 targets.

S8. `px` is a fixed size on screen. `rem` is relative to the root (`html`) font size, usually 16px, so `1.5rem` = 24px, and it grows if the visitor sets a bigger default font. A fallback (`"Momo Sans", Arial, sans-serif`) is used when the first font is not on the visitor's computer; without it, the browser falls back to its default, usually Times. See D2-10 bug 4.

S9. Parcels being shipped, from the inside out: **content** (the item) → **padding** (bubble wrap) → **border** (the cardboard box) → **margin** (the gap between parcels).

S10. `display: none` removes the element completely: it takes **no space** ("left the room"). `visibility: hidden` makes it invisible but it **still takes its space**, so you see an empty gap. See D2-HW4 bug 10.

Part 2: Interview answer key

T1. What is semantic HTML?

Say this: "Semantic HTML means using tags that say what the content *is*: `header`, `nav`, `main`, `article`, `footer`, instead of `div` for everything. It's like rooms in a house with names: kitchen, bedroom. Google, screen readers and even AI tools read those names. It doesn't change how the page looks; CSS does that. It changes what the page *means*." **They're checking:** that you know semantics ≠ styling. **Weak answer:** "HTML5 tags that

look better." (They look the same as divs.) **Likely follow-up:** "Is `<div>` bad, then?" → "No, a div is an unlabeled box. It's fine for grouping things only for styling, when no semantic tag fits."

T2. `<section>` vs `<article>`; two `<main>`s?

Say this: "An `<article>` makes sense on its own: you could post it somewhere else, like one blog post or one news item. A `<section>` is a part of a page about one topic, and it should have a heading. And no: a page has exactly one visible `<main>`, the main content. Two `<main>`s is invalid; the W3C validator reports it." **They're checking:** when *not* to use a tag (D2-02). **Weak answer:** "They're the same." **Likely follow-up:** "What goes in `<aside>`?" → "Related but not main content: a 'related posts' box, a sidebar, an ad."

T3. What does Lighthouse Accessibility check?

Say this: "It runs automatic checks in Chrome DevTools, even offline, and scores out of 100. Four things that lower it: images without `alt`, inputs without a `<label>`, links with no text at all (an icon with no name), a missing `lang` on `<html>`. Also low colour contrast, headings out of order, and an iframe without a `title`. Vague text like 'click here' is NOT caught: a human must read the links. I aim for 90 or more, and then I also do a Tab-key walk, because a tool can't check everything." **They're checking:** practical a11y experience. **Weak answer:** "It checks if the site is fast." (That's the Performance report.) **Likely follow-up:** "What can't Lighthouse catch?" → "Whether an alt text is actually *good*, or a fake div button that has a role; a human must check those."

T4. Three ways to add CSS; which wins?

Say this: "Inline with the `style` attribute (a sticky note on one shirt), internal in a `<style>` block (an outfit list for one person), and external with `<link rel="stylesheet">` (the school uniform, best for many pages). If all three set the same property with the same selector strength, inline wins, because it is the most specific. Between the `<style>` block and the linked file, the one that comes **later** in the page wins: last rule wins." **They're checking:** the cascade, not only the list. **Weak answer:** "External is always applied first." (It's about order and specificity, not type.) **Likely follow-up:** "Why avoid inline styles?" → "You can't reuse them, they are hard to override, and they mix outfit into skeleton."

T5. `class` vs `id`?

Say this: "Calling students in class: a class is 'red team', many elements can share it, and one element can have several classes. An id is 'Aarav, by name': unique, one per page. I use classes for styling, almost always, and an id for one special thing, like a `#top` anchor or a label's `for`. In CSS, `.` means class and `#` means id. Both are case-sensitive: `Card` is not `card`." **They're checking:** uniqueness and reuse. **Weak answer:** "id is for JavaScript, class is for CSS." (Partly true in practice, but not the definition.) **Likely follow-up:** "Can two elements have the same id?" → "They shouldn't; it's invalid, and HTMLHint's `id-unique` flags it."

T6. Specificity

Say this: "Specificity is how the browser picks a winner when two rules set the same property on the same element. An id beats a class, and a class beats an element. So from weakest to strongest: `p`, then `.price`, then `#menu`. Only if they are equal does the later rule win." **They're checking:** you can predict CSS instead of guessing. **Weak answer:** "The last rule always wins." (Only when specificity is equal; see C1.) **Likely follow-up:** "How do you see who won?" → "Inspect → Styles: the losing declaration is crossed out."

T7. The box model; `border-box`

Say this: "Every element is a parcel being shipped: the content is the item, padding is the bubble wrap, border is the cardboard, margin is the gap between parcels. By default, `width` is only the content, so padding and border are **added** on top: `width: 300px` with 20px padding and a 5px border is 350px on screen. With `box-sizing: border-box`, the width *includes* padding and border, so it stays 300px, which is much easier to plan. Margin is never inside the width." **They're checking:** the arithmetic. It's the #1 CSS interview question. **Weak answer:**

listing the four words without the maths. **Likely follow-up:** "How do you centre a block?" → "Give it a width and `margin: 0 auto`."

T8. block / inline / inline-block

Say this: "The queue vs the bench. A **block** element stands in a queue: each one on its own line, full width, like `div`, `p`, `h1`. **Inline** elements sit side by side on a bench, like `a`, `span`, `strong`; they take only the space of their text, and you can't push them up or down: vertical margin is ignored, and vertical padding just overlaps the neighbours. **Inline-block** sits on the bench but keeps its own size, so width, height and vertical margin work. That's why a nav uses `inline-block` links." **They're checking:** why layouts behave as they do. **Weak answer:** "Inline is small, block is big." **Likely follow-up:** "How do you remove the bullets from a nav list?" → "`list-style: none` on the `ul`."

T9. "My CSS is not working." How do you debug?

Say this: "Step by step. One: is the stylesheet loading? I check F12 → Network or Sources; a 404 or a typo in `rel` means nothing loads. Two: does my selector match? I inspect the element: if my rule is not in the Styles panel at all, the selector is wrong, maybe a missing dot or a case mismatch like `Card` vs `card`. Three: is my declaration crossed out? Then another rule won, by specificity or by coming later. Four: does it have a ⚠ warning? Then the value is invalid, like `20` with no unit, and the browser dropped it. Five: is a missing `}` or a `//` comment swallowing my rule? I look just above it." **They're checking:** a method, not luck. This answer alone impresses many interviewers. **Weak answer:** "I add `!important`." (It hides the real problem.) **Likely follow-up:** "When is `!important` OK?" → "Almost never in your own code; it's a last resort that makes the next bug harder."

C1. Predict the colour

Output: green (`rgb(0, 128, 0)`, checked in Chrome). **Why:** all three rules match the same `<p>`. `#special` is an id, the strongest, so it wins. Moving `.price` to the end changes **nothing**: "last rule wins" only applies when specificity is equal. In DevTools, `.price` and `p` are crossed out.

C2. Predict the width

Output: 350px without `border-box` (300 content + 2 × 20 padding + 2 × 5 border). **300px** with `box-sizing: border-box` (Chrome `offsetWidth`: 350 and 300). The 10px margin is outside the border in both cases, so the card takes 370px of room in its row without `border-box`. **Why:** by default (`content-box`), `width` sets only the content. This is D2-12 bug 1: three 350px cards in a 960px row don't fit, so the third wraps. (The D2-12 cards are `display: inline-block`, note 08; block boxes would stack anyway.)

C3. Find the four bugs

1. `#gg0000`: `g` is not a hex digit (0–9, a–f). The declaration is **dropped**, so the colour stays the default (Chrome: `rgb(0, 0, 0)`). Fix e.g. `#990011`. (D2-10 bug 2)
2. `1.5 rem`: a space between the number and the unit makes it invalid; **dropped** (Chrome kept the default 32px for `h1`). Fix `1.5rem`. (D2-10 bug 3)
3. `padding: 20`: no unit; **dropped** (Chrome: `0px`). Only `0` may be written without a unit. Fix `20px`. (D2-FF4, D2-HW1 bug 7)
4. `border: 2px orange`: no **style** (`solid`, `dashed`…), and the default style is `none`, so there is **no border** (Chrome: `border-style: none`, width `0px`). Fix `2px solid orange`. (D2-12 bug 4)

The missing `;` after `1.5 rem` is not a bug here: the last declaration before `}` may leave it out (it's a good habit to always write it, though).

C4. "None of my CSS works"

1. `rel="stylsheet"` is a typo, so the browser doesn't treat the link as a stylesheet and loads nothing. Fix `rel="stylesheet"`. (D2-06 bug 1)

2. `href="css/styles.css"` but the file is `style.css`: a 404 (see F12 → Network). Fix `href="css/style.css"`. (D2-06 bug 2)
3. `class="Card"` but the CSS is `.card`. Class names are **case-sensitive**, so the rule doesn't match even after the file loads. Fix `class="card"`. (D2-08 bug 1)

Fix 1 and 2 first: until the stylesheet loads, no other CSS bug is visible.

C5. Horizontal nav, reference solution

File: `index.html`

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Momo House</title>
  <link rel="stylesheet" href="css/style.css">
</head>
<body>
  <nav>
    <ul class="menu">
      <li><a href="index.html">Home</a></li>
      <li><a href="index.html">Menu</a></li>
      <li><a href="index.html">Order</a></li>
      <li><a href="index.html">About</a></li>
    </ul>
  </nav>
</body>
</html>
```

(In this test file every link points to `index.html` so the page is self-contained; in the interview, keep the four real file names.)

File: `css/style.css`

```
.menu {
  list-style: none;
  margin: 0;
  padding: 0;
  background: #14213D;
}

.menu li {
  display: inline-block;
}

.menu a {
  display: inline-block;
  padding: 10px 16px;
  color: white;
  text-decoration: none;
}

.menu a:hover {
  background: #990011;
}
```

Tested with: html-validate (0 errors) and csstree-validator (0 errors); rendered in Chrome: one navy bar, 4 white links side by side, no bullets. **What earns full marks:** `list-style: none`; `inline-block` (not `inline`) on the links so the vertical padding takes real space; the hover rule; not touching the HTML; explaining *why* while typing. **Common ways candidates fail:** forgetting `list-style: none`; `display: inline` on the `a` so the 10px padding overlaps the text above (D2-14 bug 4); writing `.menu-li` instead of `.menu li` (D2-14 bug 5); `a:hoverr` (D2-08 bug 5).

Talk-about-your-project prompts: what a good answer contains

- **One stylesheet:** "School uniform: one file, every page; a change in one place updates all pages; the browser caches it once."
- **One box in DevTools:** names all four layers with real numbers from the Computed box diagram.
- **Lighthouse:** a number, and two concrete fixes (e.g. "added labels to the contact form", "changed 'click here' to 'See the menu'").

Remember the One Rule: you may only ship code you can explain, line by line.

D2-HW4-patan-craft-shop-replay · BUGS.md

D2-HW4 Patan Craft Shop replay — answer key (BUGS.md)

File: homework/_solutions/D2-HW4-patan-craft-shop-replay/BUGS.md

Stretch homework: the rest of the Day 2 surplus bugs (a11y 1–2, media 3–4, selectors 5, units 6–8, box model 9, display 10).

Line = the line in the **starter** file. Tags: V = see it · D(HH) = HTMLHint · D(F12) = Elements panel · D(LH) = Lighthouse · D(W3C) = W3C validator · R = rule check.

#	File	Line	Bug	Fix	Tag
1	`index.html`	6	No `<meta name="description">`. Google and link previews have no summary of the page.	Add `<meta name="description" content="Patan Craft Shop: handmade pottery …">` in `<head>`.	D(LH)
2	`index.html`	4	No viewport meta. On a phone the page is shown zoomed out and tiny.	Add `<meta name="viewport" content="width=device-width, initial-scale=1.0">`.	R
3	`index.html`	28	The `<source>` is placed **after** `</video>`, so the first video has nothing to play (0:00).	Move the `<source>` line inside the `<video>`, before the fallback text.	V
4	`index.html`	33	`type="video/mp3"` is not a real type for this file. The browser skips the source, so the short video will not play (you see only the poster).	`type="video/mp4"`	V
5	`css/style.css`	9	CSS `#Header`, but the HTML is `id="header"`. Case matters: no brown header band.	`#header {`	V
6	`css/style.css`	41	`padding` with 5 values. It takes 1 to 4. Dropped: the story card text touches its border.	`padding: 10px 20px 10px 20px;` (or `10px 20px`)	V
7	`css/style.css`	28	`rgb(255, 200)` is missing the blue value. Dropped: the banner has no orange background.	`rgb(255, 200, 120)`	V
8	`css/style.css`	27	`.banner { width: 120%; }` is wider than the window, so the whole page scrolls sideways.	`width: 100%;`	V
9	`css/style.css`	40	`.card { height: 100px; }` is too short for the story, so the text spills out of the card.	`min-height: 100px;` (or delete the height)	V
10	`css/style.css`	17	`.badge { visibility: hidden; }` hides the "Closed for Dashain" badge but **keeps its space**: a big empty gap under the header. The shop is open, so the badge should leave the room.	`display: none;`	V

Bug 1 shows in Lighthouse → SEO: "Document does not have a meta description". Bug 2 is a rule check: Lighthouse 13 no longer reports a missing viewport, so test it with F12 → device toolbar (Ctrl+Shift+M): without the tag the page is drawn zoomed out on a phone.